

SQL Assignment

1. Write a query to display the first name, last name, and rental date of all customers who rented a film. Use the customer and rental tables.

USE sakila;

SELECT customer.first_name, customer.last_name, rental.rental_date

FROM customer

JOIN rental ON customer.customer_id = rental.customer_id;

2. Write a query to display the first name, last name, and address of all customers whose district is either Alberta or Texas. Use the customer and address tables.

SELECT customer.first_name, customer.last_name, address.address

FROM customer

JOIN address ON customer.address_id = address.address_id

WHERE address.district IN ('Alberta', 'Texas');

3. List all films along with their category names, including films that may not belong to any category. Then sort it by category names. Use the film, film_category, and category tables.

SELECT film.title, category.name AS category_name

FROM film

LEFT JOIN film_category ON film.film_id = film_category.film_id

LEFT JOIN category ON film_category.category_id = category.category_id

ORDER BY category_name;

4. Display the number of rentals per store. Use the rental, inventory, and store tables.

SELECT store.store_id, COUNT(rental.rental_id) AS total_rentals

FROM store

JOIN inventory ON store.store_id = inventory.store_id

JOIN rental ON inventory.inventory_id = rental.inventory_id

GROUP BY store.store_id;

5. List the names of categories that have more than 50 films. Use the category and film_category tables.

```
SELECT category.name
FROM category
JOIN film_category ON category.category_id = film_category.category_id
GROUP BY category.name
HAVING COUNT(film_category.film_id) > 50;
```

6. Write a query to display the film titles that have been rented by the customer named 'Mary Smith'. Use a subquery to get the customer ID.

```
SELECT film.title
FROM film
WHERE film.film_id IN (
    SELECT inventory.film_id
    FROM inventory
    JOIN rental ON inventory.inventory_id = rental.inventory_id
    WHERE rental.customer_id = (
        SELECT customer.customer_id
        FROM customer
        WHERE customer.first_name = 'Mary' AND customer.last_name = 'Smith'
    )
);
```

7. Write a query to display the total rental income generated by each film. Use the film, inventory, rental, and payment tables.

```
SELECT film.title, SUM(payment.amount) AS total_income
FROM film
JOIN inventory ON film.film_id = inventory.film_id
JOIN rental ON inventory.inventory_id = rental.inventory_id
JOIN payment ON rental.rental_id = payment.rental_id
GROUP BY film.title;
```

8. Write a query to display the total number of films each actor has acted in. Use the actor and film_actor tables.

```
SELECT actor.first_name, actor.last_name, COUNT(film_actor.film_id) AS total_films
FROM actor
JOIN film_actor ON actor.actor_id = film_actor.actor_id
GROUP BY actor.actor_id;
```

9. Write a query to list all rentals with a case statement that displays 'New Customer' if the customer_id is greater than 300, and 'Regular Customer' otherwise. Use the rental table.

```
SELECT rental.*,
CASE
    WHEN customer_id > 300 THEN 'New Customer'
    ELSE 'Regular Customer'
END AS customer_type
FROM rental;
```

10. Write a query to find all films with titles that start with the letter 'A'. Display the film title and the release year. Use the film table and inventory table.

```
SELECT film.title, film.release_year
FROM film
WHERE film.title LIKE 'A%';
```

- 11. Create a new database and perform the following operations using SQL commands:**

- Create a table with 2 columns: `Student_ID` and `Marks`.
- Insert 2 records into this table.
- Rename the `Marks` column to `Score`.
- Delete the table.
- Delete the database.

```
CREATE DATABASE StudentDB;
USE StudentDB;
CREATE TABLE Student (
```

```
Student_ID INT,  
Marks INT  
);  
INSERT INTO Student (Student_ID, Marks) VALUES (1, 85), (2, 90);  
ALTER TABLE Student RENAME COLUMN Marks TO Score;  
DROP TABLE Student;  
DROP DATABASE StudentDB;
```